

NATIONAL TEACHERS COLLEGE
School of Arts, Sciences and Technology**STUDENT DOCUMENTATION WORKSHEET****NTC_CC105 – Data Structure w/Lab | Terminal Assessment, Weeks 1–5**

“Smart Campus Student Resource Management System”

This worksheet accompanies the NTC_CC105 Terminal Assessment (Weeks 1–5). Complete every section as a group, then submit this worksheet together with your source code, GitHub repository, and screenshots. Refer to the Terminal Assessment document for full requirement details.

Group No.: _____**Section:** BSIT 2.4**Date:** September 20, 2026**Instructor:** Prof. Darwin C. Vargas, MIT**Group Members and Assigned Positions**

No.	Full Name	Assigned Position
1	Mier, Ronamae	Project Lead / System Integrator
2	Quintela, Nicole M.	Data Structure Developer
3	Serdenia, Sheayla Rihanna A.	Memory & Pointer Developer
4	Cunanan, Jhoey	Matrix & Algorithm Developer
5	De Pedro, Precious Mae M.	Documentation & QA Lead

SECTION 8 – TEST CASES (Part IX, minimum of 10)

No.	Feature	Input	Expected Output	Actual Output	Status
1	Add Student	ID: 101 Name: Alice Section: 1.1 Year level: 1st Year	Student added successfully.	Student added successfully.	Passed
2	Add Student	ID: 101 Name: Alice Section: 1.1 Year level: 1st Year	System rejects with an error message	Error: Student ID 101 already exists.	Passed
3	Display Student	Non-empty array	All active records listed	Shows all the active student Records	Passed
4	Search Student	ID:101	Matching record displayed	Match found.	Passed
5	Search Student	ID: 203	"Record Not Found" Message	Record not found.	Passed
6	Update Student	ID: 101 New Section: 1.2 New Year Level: 2nd Year	Record fields updated	Student updated successfully	Passed
7	Deletion Student	ID:101	Record removed, array reindexed	Student deleted successfully.	Passed
8	Bounds Checking	Index beyond array capacity	System blocks and reports invalid index	Error: Cannot add student. Student array is full.	Passed
9	Maximum Capacity	Adding More when the array is full.	System reports array full, add rejected	Error: Cannot add student. Student array is full.	Passed
10	Matrix Processing	Sample resource matrix	Correct row/column totals displayed	Department totals: D1: 28, D2: 25, D3: 23, D4: 21 Resource totals: R-01: 30, R-02: 19, R-03: 28, R-04: 20	Passed